

Macroinvertebrate communities of non-glacial high altitude intermittent streams

M. L. CHAVES*, M. RIERADEVALL[†], P. CHAINHO*, J. L. COSTA*, M. J. COSTA*[‡] AND N. PRAT[†]

*Universidade de Lisboa, Faculdade de Ciências, Instituto de Oceanografia, Lisboa, Portugal

[†]Departament d'Ecologia, Universitat de Barcelona, Barcelona, Espanya

[‡]Universidade de Lisboa, Faculdade de Ciências, Departamento de Biologia Animal, Lisboa, Portugal

SUMMARY

1. Macroinvertebrate assemblages of five non-glacial intermittent high altitude headwater streams (above 1400 m – Serra da Estrela, Portugal), with dry periods of different lengths (0–3 months), were investigated in nearly undisturbed conditions to (i) examine spatial differences and identify environmental variables responsible for the observed invertebrate patterns, (ii) assess the association of dry period length with invertebrate community structure and (iii) determine the influence of using different taxonomic identification levels (order, family and genus) to assess invertebrate community patterns.
2. More than 100 macroinvertebrate genera were identified. Insects clearly dominated these communities with more than 95% of total captures and around 95% of the total richness. Diptera were the most rich and abundant group with chironomid occurrences comprising over 70% of macroinvertebrate captures.
3. The highest taxon richness, diversity, EPT (Ephemeroptera + Plecoptera + Trichoptera) and OCH (Odonata + Coleoptera + Heteroptera) genus richness, the greatest number of exclusive and characteristic taxa identified by the Indicator Value (IndVal), and a distinct community structure shown by Canonical Correspondence Analyses (CCA), were found in the only stream that was never totally dry, with pools lasting over summer. Environmental gradients that spatially structured the macroinvertebrate communities were always related to flow variations.
4. Over time, the highest abundances found in these systems were also related to flow variations and maximum genus richness occurred in the connected pools or in isolated pools. Streams with longer dry periods presented a distinct recolonization phase, with higher abundance of the stonefly larvae *Nemoura* sp. and the presence of the chironomid larvae *Krenosmittia* sp., possibly arriving from the hyporheos.
5. Taxonomic level of invertebrate identification was vital for recognizing the characteristic taxa (IndVal) of streams yet was not critical for identifying streams with the highest macroinvertebrate richness/diversity or structuring environmental gradients.
6. Overall, this study emphasizes the variability of high altitude intermittent streams macroinvertebrate communities, despite spatial proximity. This variability was probably related to flow intermittency and hydrologic permanence, different vegetation covers and riverbed substrata. Consequently, the establishment of reference conditions should involve long-term data collections and more detailed physical characterization. Also, these

Correspondence: M. L. Chaves, Universidade de Lisboa, Faculdade de Ciências, Instituto de Oceanografia, Campo Grande, 1749-016 Lisboa, Portugal.

E-mail: mlchaves@fc.ul.pt

findings have significant implications for accurately predicting the ecological consequences of future climate change in high altitude scenarios.

Keywords: dry period, headwater streams, natural variability, taxonomic levels, undisturbed systems

Introduction

Europe has warmed more than the global average during the 20th century and the greatest warming has been recorded in northwest Russia and the Iberian Peninsula (European Environment Agency, 2004). Mountainous regions are particularly vulnerable to adverse effects of climate change (Allan & Flecker, 1993) and are already suffering from higher than average increases in temperature (European Environment Agency, 2006). High mountains on the Iberian Peninsula will probably be among the first areas in Europe showing significant alterations in riverine fauna produced by climate change and the assessment of future change requires a good knowledge of the pristine condition or at least of the current situation.

Simultaneously, over the last 5 years growing interest in pristine aquatic systems has been raised by the Water Framework Directive requirements (WFD; European Commission, 2000). However, this recent European Directive, which among other things aims to protect inland surface water, considers only streams with a catchment area larger than 10 km² (WFD-System A descriptors). Consequently, many small high altitude lotic systems will not be included in future monitoring and protection actions. It is clear that no protection law can succeed without guaranteeing the functionality of these headwater streams as the health and productivity of streams, rivers, lakes and estuaries depend upon the essential ecosystem services provided by these intact systems. These services include mitigating floods, maintaining water quality and quantity, recharging groundwater, trapping sediments, recycling nutrients, providing habitat for biota and supplying food resources to downstream ecosystems (Meyer *et al.*, 2003). The estimation of the extent of these small streams and therefore their real proportion and importance in the stream network is hindered by the fact that they are often unnamed and rarely appear on maps (see Meyer *et al.*, 2003 and Fritz, Johnson & Walters, 2006).

Regardless of climatic region and altitude, one of the most distinctive and ecologically influential characteristics of many headwater streams is natural drying (Fritz *et al.*, 2006). Within alpine areas, streams and rivers show different hydrological characteristics, depending on the prevailing water sources and regional climate (Füreder *et al.*, 2001). Under Mediterranean-climate conditions, most streams are characterized by sequential floods and droughts that are seasonally predictable but show variable intensity (e.g. Gasith & Resh, 1999) and can be typified as seasonal droughts (Lake, 2003). Temporary streams (e.g. intermittent, ephemeral) have recurrent dry phases (Uys & O'Keeffe, 1997; Fritz *et al.*, 2006; Williams, 2006). However, complex hydraulic characteristics and mean velocity are among the features that best explain the distribution of macroinvertebrates (e.g. Statzner, Gore & Resh, 1988; Chaves *et al.*, 2005) and therefore the drying of an aquatic habitat represents a significant natural disturbance for organisms living there (Delucchi, 1988). Although seasonal droughts are press disturbances (see Lake, 2003), the ecological response might occur as irregular steps (Boulton, 2003; Acuña *et al.*, 2005) or following a ramp model (Lake, 2000, 2003; Bonada *et al.*, 2006). In some situations, the compositions of the benthic invertebrate communities before versus after the dry period (e.g. Ladle & Bass, 1981) and during flow versus pool conditions (Rabeni, Doisy & Galat, 2002) are similar, but most frequently benthic communities are different, especially under Mediterranean-climate conditions (e.g. Rieradevall, Bonada & Prat, 1999; Bonada *et al.*, 2006).

Remote mountain lakes and streams are already affected by numerous human activities at local (water abstraction and tourism) and regional scales (climate change and acidic rains) (e.g. Camarero *et al.*, 1995; McGregor *et al.*, 1995), although much less impacted than lowland running waters (Brittain, Lencioni & Maiolini, 1998; Ward, 2002). Species occurring in intermittent headwater streams can be very sensitive to natural and anthropogenic altera-

tions (i) that increase aridity, such as climate change and water abstraction, as these changes might alter the way in which assemblages respond to drought (Bonada *et al.*, 2006), but also to those (ii) that modify the surrounding land (vegetation clear-cutting and urbanization), due to the close terrestrial-aquatic connection present in very small streams (Lowe & Likens, 2005). High altitude aquatic systems are not only vulnerable to environmental alterations due to their sensitivity, but also excellent indicators of such changes (McGregor *et al.*, 1995; Rieradevall, Jimenez & Prat, 1998). To be used as indicators, reference conditions should be established for high altitude intermittent streams. However, information on taxonomy and ecology of aquatic invertebrates of these systems, particularly non-kryal streams, is still scarce (Maiolini *et al.*, 2006). This is especially true for the westernmost high mountains of continental Europe, the Serra da Estrela (Portugal). In Portugal, studies on invertebrate communities of high altitude freshwater systems have been largely ignored and, to our knowledge, besides some occasional entomological visits focussed on a specific taxon (e.g. Cobo, Soriano & González, 2001) and the study of some lagoons (Rieradevall *et al.*, 1998; Rieradevall & Prat, 1999), no further work has been carried out. This high altitude region can be taxonomically rich in some macroinvertebrate groups. Thus, a second problem in the establishment of reference conditions could be how to select the appropriate taxonomic resolution as genus/species identification may actually reduce the ability to detect a deviation from the reference condition (Bailey, Norris & Reynoldson, 2001). The main objective of this study was to characterize benthic macroinvertebrate communities of intermittent high altitude streams (above 1400 m a.s.l.) in Serra da Estrela (Portugal), by (i) determining if there are differences among the invertebrate community structure of five proximal streams and identifying environmental variables responsible for the observed patterns of invertebrate distribution, (ii) investigating if similar temporal flow regimes (length of the dry period) display similar community structures and (iii) verifying if information obtained with the invertebrate community was strongly changed by the use of three different taxonomic identification levels (order, family and genus).

Methods

Study area

Benthic invertebrates and associated environmental data were collected in five different headwater streams located within the Mondego River basin (Fig. 1), in Serra da Estrela Natural Park (PNSE). The central massif of Serra da Estrela, where this study focuses, is characterized by granitic geology, with a maximum elevation of 1993 m above sea level (a.s.l.). PNSE represents a well preserved area of Serra da Estrela, with 0.2 % of its area covered by aquatic systems, 1.5% by artificial areas (urban, industrial, roads) and 2% and 23% of the park is used for intensive and extensive cropping respectively. More than 70% of the PNSE region includes natural and semi-natural areas. Climate data available for Serra da Estrela, measured at approximately 1380 m a.s.l., indicate a quite predictable Mediterranean climate, with warm, dry summers and cold, wet winters (Vieira, Mora & Ramos, 2003). In general, the wet season extends from October to May (Daveau *et al.*, 1977). The mean annual rainfall and temperature are 1692.2 mm and 8.9 °C, respectively (Lima & Lima, 2002). On average, the coldest month is January (2.8 °C), while the warmest is July (17.1 °C) and the wettest and the driest months are December (253.8 mm) and August (0.1 mm) respectively (Lima & Lima, 2002). Snowfall data show a median of 40 to 50 days per year at 1400–1600 m a.s.l. (Andrade *et al.*, 1992), but snow presence is almost constant from November to April. Several small watercourses are covered by ice through the winter, while during summer many streams are intermittent due to low precipitation. The high plateaus of Serra da Estrela were covered by an ice-cap with around 70 km² during the last glaciation (Daveau, 1971).

The five headstreams chosen for this study were located at altitudes above 1400 m a.s.l. and between 40°25.700 and 40°21.445N and 7°37.847 and 7°35.172W (Table 1). Distance between sites ranged from 16 km to 0.5 km. Therefore, all sites presented very similar climatic and geological conditions, conforming to a single stream type (>800 m a.s.l., siliceous geology, with a catchment area less than 10 km²).

These streams can have two seasonal discharge peaks: one due to winter precipitation and the other

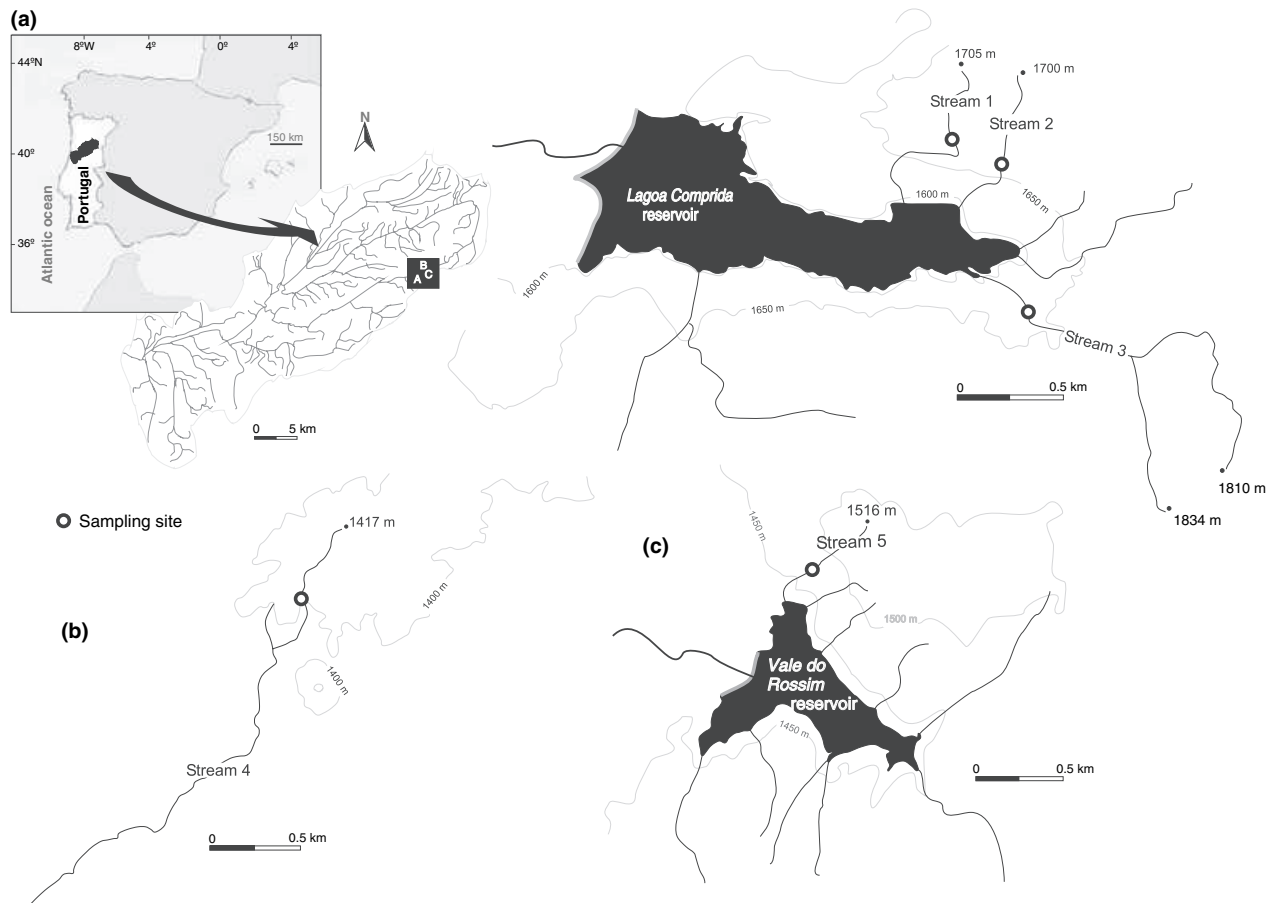


Fig. 1 The location of sampling sites within the Mondego River basin is indicated in the small black square. Detailed geographic position of each stream in the areas A, B and C is also shown.

to melting ice and snow during spring. All studied streams were *intermittent seasonal* (Uys & O’Keeffe, 1997) and sites were classified according to temporal flow regime as (i) never dry although forming pools (Stream 3), (ii) dry for less than 3 months (Streams 1 and 2) and (iii) dry for at least 3 months (Streams 4 and 5) (Fig. 2). Streams 1, 2 and 3 feed the Lagoa Comprida reservoir and were already covered by ice during late autumn, while stream 5 drains into Vale do Rossim reservoir (Fig. 1). The shrub *Erica* sp. dominated the riparian areas of streams 1, 2 and 3 and no trees were present; stream 4 had no tree or shrub riparian vegetation, while deciduous trees prevailed at stream 5, consisting mainly of *Acer* sp. Stream 4’s catchment area had the occasional presence of livestock during spring and summer, stream 1 was partially affected by fire during summer 2003 (before the first sampling campaign), whereas

stream 5’s catchment area is located near a camp site opened only during summer. In general, all sites were roughly undisturbed systems as they are inside a protected area, free of point-source pollution, with non-existent or minimal agricultural runoff in their catchments and away from other human activities.

Sampling

The five stream sites were sampled on six occasions (Fig. 2), at approximately 4–6-weekly intervals, from late April 2004 to early November of the same year. The sampling period was constrained by inaccessibility during the snow and ice season. Four of the five sampling sites were completely dry for at least one sampling occasion (Fig. 2). Sites were sampled only when water was present.

Table 1 Environmental characteristics of each sampling site in the five studied streams in Serra da Estrela. Mean and ranges of values (in brackets) are given for variables that changed among sampling occasions

Variables	Streams				
	1	2	3	4	5
Latitude (North)	40°21.840	40°21.775	40°21.445	40°25.700	40°24.244
Longitude (West)	7°37.847	7°37.713	7°37.630	7°35.653	7°35.172
Distance from source (km)	0.38	0.40	2.00	0.73	0.50
Altitude (m a.s.l.)	1673	1664	1629	1400	1470
Drainage basin (km ²)	0.08	0.15	2.16	0.20	0.13
Mean channel width (cm)	43.0 (30–55)	42.4 (20–57)	148.0 (90–350)	25.0 (10–40)	33.3 (25–50)
Maximum depth (cm)	19.8 (4–30)	30.2 (6–50)	49.3 (30–100)	13.3 (10–20)	12.7 (1–20)
Discharge (L s ⁻¹)	3.57 (0.02–10.98)	1.76 (0.02–7.79)	131.90 (3.40–724.75)	3.85 (0.10–6.12)	1.28 (0.03–3.30)
Water temperature (°C)	13.9 (2.1–26.4)	10.4 (4.0–19.4)	12.9 (2.6–18.5)	8.0 (6.2–9.8)	7.3 (6.1–8.3)
pH	5.95 (5.47–6.45)	5.68 (5.08–6.26)	5.71 (5.21–6.00)	5.58 (5.17–6.00)	5.48 (5.45–5.50)
Dissolved oxygen (mg L ⁻¹)	8.45 (7.37–9.37)	8.87 (8.55–9.43)	8.23 (6.05–9.86)	7.67 (6.44–8.62)	7.67 (7.07–8.71)
Conductivity (µS cm ⁻¹ at 25 °C)	22 (8–46)	14 (7–24)	17 (9–25)	18 (11–24)	16 (9–22)
Total phosphorus [†] (mg L ⁻¹)	<0.06	<0.06	<0.06–0.55	<0.06	<0.06
Stressor source (reduced)	Natural	Fire	Natural	Livestock	Camp site
Dominant substratum class	Bedrock	Bedrock	Bedrock	Gravel	Gravel
Sediment free fraction [‡]	Sand, gravel, cobble, silt	Gravel, sand, silt	Gravel, sand, silt	Gravel, sand, cobble, boulder, silt	Gravel, sand, boulder, silt
Total organic content (%) [*]	7.5	0.9	1.5	0.9	4.3

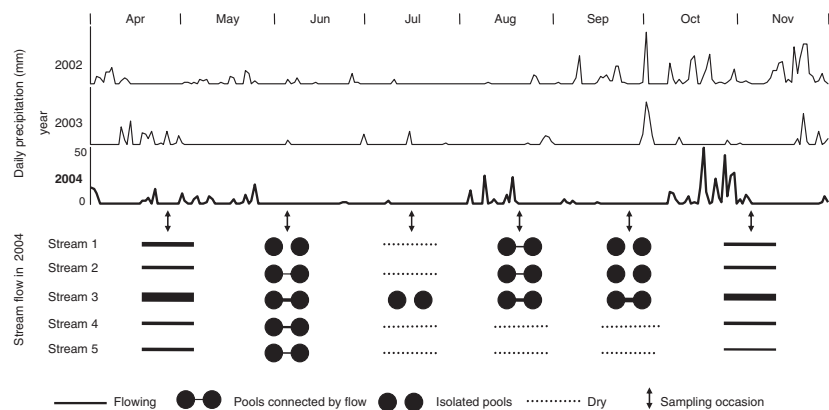
Streams 1 and 2 were sampled on five occasions, stream 3 on six and streams 4 and 5 on three.

*Indicates environmental variables measured only once.

[†]All five streams had non-detectable water concentrations of nitrates (<0.05 mg L⁻¹), nitrites (<1.0 mg L⁻¹) and ammonia (<0.08 mg L⁻¹).

[‡]Substratum size classes, excluding bedrock, are ordered by dominance at each site.

Fig. 2 Flow conditions in each stream during sampling occasions; line thickness is proportional to relative discharge: thicker line corresponds to 724.75 L s⁻¹, and thinner line corresponds to 0.02 L s⁻¹. Daily precipitation values during the study period and during previous years in Serra da Estrela are indicated on top graphics (source: <http://www.meteo.pt>). Sampling occasion dates are indicated by arrows.



Macroinvertebrates

Macroinvertebrates were always collected by the same operator using a 30 × 30 cm kick-net with a 250 µm mesh size. Care was taken to include all possible habitats over representative sections of the stream (10 m samples), similar to the multi-habitat sampling procedure used in the USA (Barbour *et al.*, 1999). Effort was allocated in proportion to the occurrence of each habitat. The sorting process of a

biological sample followed three sequential steps: first, all specimens larger than 1000 µm were sorted out; second, at least 250 specimens (<1000 µm) were removed according to an area-based method to allow an extrapolation of macroinvertebrate total abundance; finally, all the sample was screened for non-abundant specimens, i.e. taxa which had not been sorted into the 250 specimens-group, to guarantee that all different taxa were detected. Macroinvertebrates were identified mostly to the genus level (whenever

possible) but also to higher and lower taxonomic levels (classes or species). Microscope slide preparation of Chironomidae larvae involved a sequential procedure of clearing individuals with a 10% KOH solution for 15–25 min (depending on individual size) at 85 °C. This procedure was followed by neutralization at room temperature, initiation of dehydration with 70% ethanol, then mounting from 90% ethanol into Euparal.

Abiotic variables

On each sampling occasion, physical and chemical variables were measured at each site prior to invertebrate fauna sampling. Water temperature (°C), dissolved oxygen (mg L⁻¹), conductivity (µS cm⁻¹), total dissolved solids (TDS mg L⁻¹), pH and salinity were measured in situ with a Data Sonde Surveyor 4 portable probe (Yellow Springs, Ohio, USA). Water samples were collected below the water surface in the central region of water courses, frozen, and analyzed with certified methods (ISO/IEC 17025) to determine concentrations of nitrates (NO₃⁻ mg L⁻¹), nitrites (NO₂⁻ mg L⁻¹), total phosphorus (P mg L⁻¹) and ammonia (NH₄⁺ mg L⁻¹). Discharge (L s⁻¹) was determined in the narrowest section of the stretch by dividing the total channel width into, at least, seven equal sections and measuring the channel depth and current velocity (using a Hydro-Bios instrument; RHCM, Kiel-Holtenau, Germany) in each section; discharge was calculated by integrating medium depth (m), total width (m) and current velocity (m s⁻¹). Substratum samples were taken in the same reach on a single sampling occasion (November), to estimate sediment grain size and total organic content (TOC). After ignition at 550 °C, six sediment grain size classes were estimated by sieving the substratum sample through an AFNOR-type sieve battery (CISA, Barcelona, Spain – 9.25, 2.00, 0.50, 0.25 and 0.063 mm) and weighing each fraction. TOC was obtained by deducting the ash weight, measured after ignition at 550 °C for 6 h, from the dry weight, measured after drying at 60 °C for 24 h. Dominant substratum class was visually estimated on every sampling occasion using five different groups: (i) silt and clay; (ii) sand; (iii) pebbles and gravel; (iv) cobble, boulders and stones and (v) bedrock. Altitude (m), latitude and longitude (degrees° minutes' seconds'') were registered with a GPS system at each site. Drainage basin

area and distance from source were calculated from digitized 1 : 50 000 scale maps. Data of daily rainfall over the studied period and previous years were available from Seia climate station (Fig. 2).

Data analysis

A Principal Components Analysis (PCA) was performed to highlight similarities and differences of abiotic conditions between streams and within streams during a range of sampling occasions, using the software CANOCO for Windows ver. 4.5 (ter Braak & Šmilauer, 2002). Highly correlated (Pearson, $P < 0.01$) environmental variables were removed from the PCA.

Macroinvertebrate abundance (N), taxon richness (S) and Shannon-Wiener's diversity index (H, natural log) were calculated as descriptive measures of the benthic community for each site and sampling occasion (Legendre & Legendre, 1976). The relationship between the indices EPT (Ephemeroptera, Plecoptera and Trichoptera richness- taxa more related to riffle areas) and OCH (Odonata, Coleoptera and Heteroptera richness- taxa representative of lentic habitats) reflects the temporariness of a stream as demonstrated by Bonada *et al.* (2006) among others. Thus EPT relative to EPT + OCH [EPT/(EPT + OCH)] and Diptera + Oligochaeta abundances (taxa normally associated with extreme environments) relative to total captures [(Diptera + Oligochaeta)/Total] were analysed among streams and sampling occasions to investigate changes in the invertebrate community between streams and within streams through time. A Generalized Least Squares (GLS) model, using the *gls* function in the *nlme* package in software R (R Development Core Team, 2006), was used to test if mean taxon richness, diversity, abundance and OCH, EPT and Diptera genus richness were equal among streams and among flowing conditions and pools (isolated and connected). Small sample size precluded the estimation of interaction effects. Since streams were repeatedly sampled over time, temporal autocorrelation was permitted in model residuals. Specifically, an AR(1) correlation structure was specified in the GLS models. To test for the significance of each model factor, a GLS based ANOVA (Analysis of Variance) table was used.

Ordination models were performed with the software CANOCO for Windows ver. 4.5 (ter Braak &

Šmilauer, 2002) to analyse spatial variations in taxon abundance and to identify relations between environmental (25 variables) and biological data. The ordination model used in each analysis was selected based on the length of the gradient calculated by a Detrended Correspondence Analysis (Jongman, ter Braak & van Tongeren, 1995). Canonical Correspondence Analysis (CCA) was used since the gradient length of the first axis was always between 3 and 4 units of mean standard deviation of species turnover. Three CCAs were performed to investigate which environmental variables were correlated with spatial macroinvertebrate community patterns, considering three levels of taxonomic identification (13 major groups, mostly orders; 48 families; and 129 genera). Biological abundance data were log (x+1)-transformed prior to all ordination analyses and no taxa removal was performed. Seasonal environmental parameters and macroinvertebrate abundance matrices were reduced to average values per stream. Since the number of environmental variables retained in each analysis should be considerably less than the number of samples included, a forward manual selection of standardized (mean 0 and variance 1) environmental variables was performed, using a Monte Carlo permutation test (9999 unrestricted permutations; $P < 0.05$) (ter Braak & Verdonschot, 1995). The significance of the first axis and trace (sum of all canonical eigenvalues) were tested for each CCA with a Monte Carlo permutation test (9999 unrestricted permutations, $P < 0.05$) (ter Braak & Šmilauer, 2002). Preliminary Correspondence Analyses (CA) were performed to check for spatial patterns without being constrained by environmental variables.

The Indicator Value (IndVal, Dufrêne & Legendre, 1997) method was used to determine the most representative macroinvertebrate taxa in each of the naturally occurring groups (streams 1, 2, 3, 4, and 5, with sampling occasions as replicates). The IndVal method determines indicator taxa by combining the relative taxon abundance with its relative frequency of occurrence in the various groups of sites. The IndVal for each taxon ranges from 0% to a maximum value of 100%, which is attained when all specimens of a taxon are found in a single group of samples and when the taxon occurs in all samples of that group. Taxa with an IndVal higher than 25% (i.e. with specimens present in more than 50% of the samples of one group and with a relative abundance

in that same group of more than 50%) contribute to the habitat specificity and only these were considered as characteristic taxa. To investigate if a taxon was characteristic of a stream the significance ($P < 0.05$) of each taxon IndVal was tested by a site randomization procedure that reallocates samples among sample groups (9999 permutations) (Dufrêne & Legendre, 1997). Thus, an IndVal and a P -value were associated to each taxon. The IndVal2 package (Dufrêne & Legendre, 1997) was used for the computations.

Results

Environmental characterization

Annual rainfall data in the region showed high variability among different years (Fig. 2). However, summer months are always the driest with the rainy season beginning in late summer or during autumn (Fig. 2). Discharge was highly variable during the sampling period in all streams, but stream 3 showed the highest variation (Table 1, Fig. 3) and was the only stream with water for the entire year (Fig. 2). The

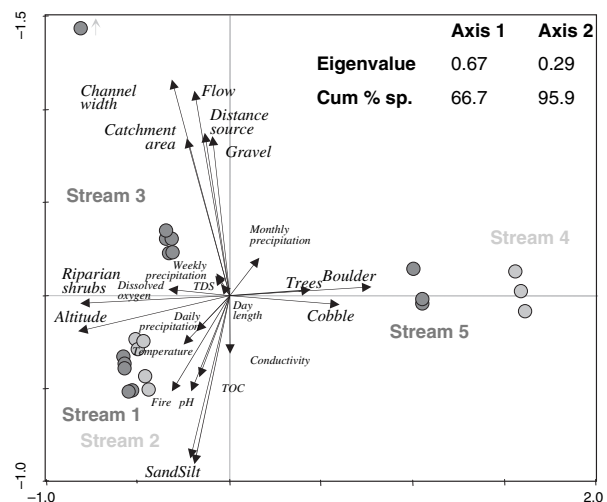


Fig. 3 Diagram of Principal Component Analysis of environmental variables. Horizontally first axis, vertically second axis. Sampling sites are represented by circles (small arrow next to a site- for stream 3 in April- indicates that the true y value is outside the plot) and environmental variables by solid arrows. The length of the arrow is a measure of the importance of the variable and the arrowhead points in the direction of increasing influence. Maximum depth and salinity were highly correlated with channel width ($R = 0.84$; $P < 0.01$) and conductivity ($R = 0.75$; $P < 0.01$), respectively, and are not plotted.

greatest abiotic similarities were found between streams 1 and 2 while streams 3, 4 and 5 differed from each other. Although the resemblance among streams 4 and 5 seemed strong, this was mainly due to altitude and flow conditions (Fig. 3, Table 1). Abiotic differences through the year within the same site were very small except for the April sampling campaign in stream 3 (Fig. 3). Low flows or isolated pools during summer were generally associated with warmer water, lower dissolved oxygen, increased concentrations of ions (higher conductivity) and slightly higher pH, although always below neutrality. Streams 4 and 5 presented the smallest amplitude of values for the majority of abiotic variables, since they were dry during summer months (Table 1). Stream 3 had the highest habitat heterogeneity with some soft bottom pools, besides the granite bedrock pools, and was the only one containing macrophytes. Mosses were present in all streams, although rare in stream 4.

High altitude macroinvertebrate communities

Macroinvertebrate cumulative taxon richness indicated that 13 major groups (class and order), 45 families and more than 100 genera were identified in the 22 samples collected in the five studied streams (Appendix).

Insects clearly dominated these high altitude benthic communities with more than 95% of total captures and around 95% of total richness (Tables 2 & 3). However, the Oligochaeta class (Annelida) was the fifth most abundant group. Diptera were the most rich (Table 3) and abundant group due to predominant chironomid genera occurrences through the entire year (Table 2, Fig. 4). The driest sampling occasion (July) was an exception as the invertebrate community, reduced to stream 3, was dominated by a mayfly, *Habrophlebia* sp. (Fig. 4). Although the number of genera included in each insect order ranged from 4 to 48 (Table 3), only one or two genera were responsible for most of the total abundance of an insect order (Fig. 4).

Macroinvertebrate spatial distribution

Significant differences were found for taxon richness (GLS-based ANOVA: order: $F = 67.96$; family: $F = 114.36$ and genus: $F = 12.32$; d.f. = 4, 15; $P < 0.05$) and diversity (GLS-based ANOVA: order: $F = 3.64$; family: $F = 3.01$ and genus: $F = 9.70$; d.f. = 4, 15; $P < 0.05$) among the five streams. Macroinvertebrate taxon richness and diversity had the highest average values in stream 3 (Fig. 5a,b). No statistical differences were found for invertebrate abundances among streams (GLS-based ANOVA: $F = 0.29$; d.f. = 4, 15; $P = 0.88$) and abundance was highly variable within each stream, particularly in streams 1, 2 and 3 (Fig. 5c). Significant differences were found for OCH and EPT genus richness among streams (GLS-based ANOVA: $F_{\text{OCH}} = 5.05$ and $F_{\text{EPT}} = 6.01$; d.f. = 4, 15; $P < 0.05$), but not for Diptera genus richness (GLS-based ANOVA: $F_{\text{Dipt}} = 1.78$; d.f. = 4, 15; $P = 0.78$). Molluscs, leeches and OCH genus richness, especially coleopterans, were much higher in stream 3 (Fig. 5d). Cumulative taxon richness was greater than 10 orders, 30 families and 80 genera in stream 3, whereas in the other streams these values were respectively fewer than 10, around 20 and fewer than 50 (Table 4).

Canonical Correspondence Analysis results show that stream 3 was clearly separated from the other

Table 2 Mean abundance with standard deviation and relative percentage of the main groups found in 22 samples collected from five high altitude streams in Serra da Estrela

Invertebrate groups	Mean abundance	Standard deviation	Relative abundance (%)
EPT	183.04	154.60	17.28
OCH	62.04	150.19	5.86
Diptera	770.32	819.08	72.74 (70.23)
Annelida	41.91	60.78	3.96
Mollusca	1.50	3.16	0.14
Others	0.23	0.61	0.02

Chironomidae relative abundance is given in brackets.
EPT – Ephemeroptera, Plecoptera, Trichoptera; OCH – Odonata, Coleoptera, Heteroptera; Annelida – Oligochaeta and Hirudinea; Others – Lepidoptera, Hydracarina, Tricladida.

Table 3 Insect taxon richness found in 22 samples from five high altitude streams in Serra da Estrela for each insect order

Insect orders	Dip	Ple	Col	Eph	Tri	Het	Odo	Total
Family	12	4	9	3	6	4	3	41
Genus	48	5	25	5	11	6	4	104

Taxon abbreviations are defined in Table 6.

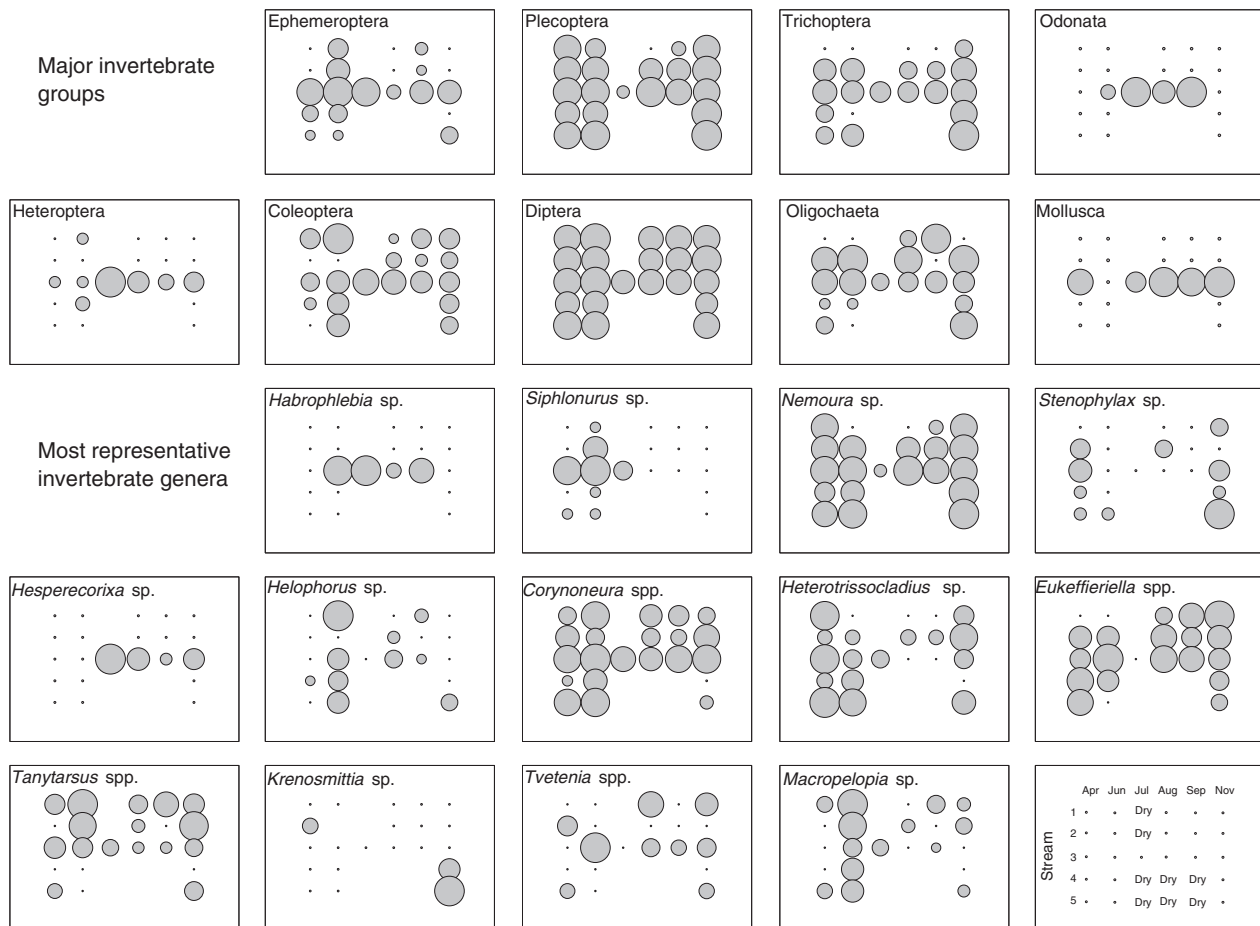


Fig. 4 Spatio-temporal distribution of 22 taxa recorded in five high altitude streams of Serra da Estrela, on six sampling occasions. The size of the circles is proportional to taxon abundance (log x -transformed). Major macroinvertebrate groups that accounted for at least 0.1% of total captures in Serra da Estrela, most representative genera of each invertebrate group, and genera that accounted for more than 100 specimens in a single sampling occasion are illustrated. The bottom-right square indicates streams and sampling campaigns.

streams along the first canonical axis in all CCA taxonomic models (Fig. 6). In these constrained ordinations all first canonical axes were statistically significant (F -ratio_{order} = 6.198, F -ratio_{family} = 2.565, F -ratio_{genus} = 1.652; $P < 0.05$). About 83.1%, 70.9% and 63.2% of the total variance in taxon abundance was accounted by the first two ordination axes in the order, family and genus CCAs respectively. All CCA models retained a single significant environmental variable: (i) discharge for order level data and (ii) channel width for family and genus data. Therefore, abiotic gradients established along first ordination axes were always related to flow variations (Fig. 6).

Of the total taxa occurring in these high altitude streams (most to genus level), 27.9% were exclusive to

stream 3, accounting for 3.7% of total abundance (Table 5). Only five genera of these 36 exclusive taxa were captured in more than half (>3) of sampling occasions in stream 3. Five major groups, seven families and nine genera had an IndVal significantly higher than 25% ($P < 0.05$) and were therefore considered as taxa characteristic of stream 3 (Tables 6–8). In this stream, the leeches Erpobdellidae (*Dina* sp.) and the mayflies Leptophlebiidae showed an IndVal of 100% (Tables 6–8), indicating that these taxa were exclusive to this stream and occurred on all sampling occasions, being the most characteristic taxa of stream 3 invertebrate community. Though with an IndVal much lower than 100%, the mayfly genus *Habrophlebia* sp. alone, was still considered significantly characteristic of the same stream (Table 8,

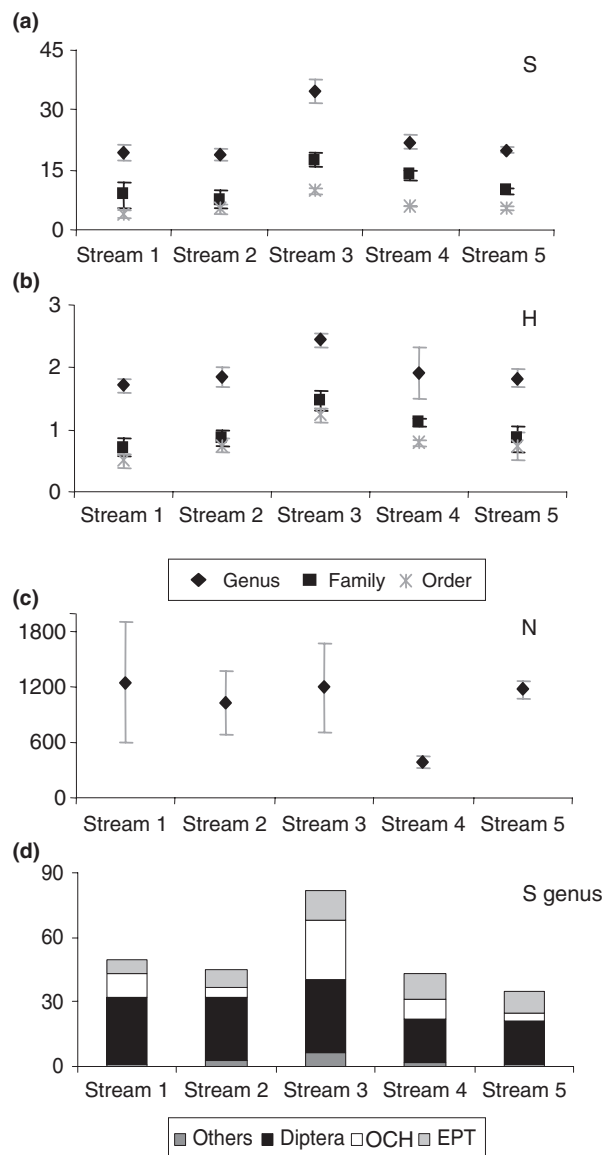


Fig. 5 Average taxon richness (plot a) and Shannon-Wiener diversity (plot b) for genus, family and order in each stream and respective standard errors. Average abundance for each stream and respective standard errors are shown in plot c. Numbers of genera for EPT, OCH and Diptera are given in plot d. Others include Hirudinea, Oligochaeta, Lepidoptera, Tricladida, Hydracarina and Mollusca.

Table 4 Cumulative number of taxa per stream in Serra da Estrela

Taxonomic level	Stream				
	1	2	3	4	5
Order	7	8	11	8	6
Family	19	19	32	22	16
Genus	50	45	82	43	35

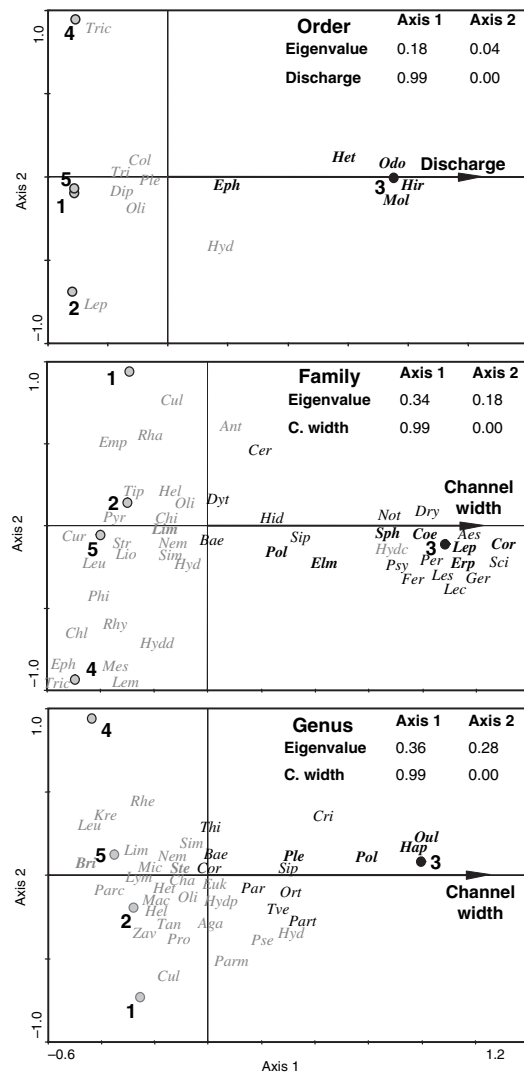


Fig. 6 Diagrams of Canonical Correspondence Analysis (CCA) of taxa invertebrate abundance divided into (a) major groups biological data (mostly orders) of 22 samples with one significant environmental variable ($P < 0.01$); (b) family biological data of 22 samples with one significant environmental variable ($P < 0.05$) and (c) genus biological data of 22 samples with one significant environmental variable ($P < 0.05$). Genus CCA was performed with the total matrix of genera, but genera with less than 15% of weight range were removed from the plot. See Tables 6–8 for order, family and genus abbreviations. Other abbreviations are: Bae, Baetidae n.i.; Lim, Limoniidae; Lym, *Lymnophyes* sp.; Ort, *Orthocladus* spp.; Par, *Paraphaenocladus* sp.; Parm, *Paramecina* sp.; Rhe, *Rheocricotopus* sp.; Sim, *Simulium* sp. Taxa in black are related to stream 3, when in bold are significantly characteristic of stream 3 according to IndVal results. Sampling sites are represented by circles and significant environmental variables by solid arrows. The length of the arrow is a measure of the importance of the variable and the arrowhead points at the direction of increasing influence. Correlations values of environmental variables with first two axes and the first two eigenvalues of each ordination analysis are also given.

Fig. 4). Heteropterans occurred in three streams but in two their presence was highly accidental, with just one specimen in stream 1 and two specimens in stream 4 (Fig. 4). Consequently, the highly significant IndVal on Table 6 identified the order Heteroptera as a taxon characteristic of stream 3. Within heteropterans, the Corixidae family and the *Hesperocorixa* sp. genus were also characteristic of the same stream (Tables 7 & 8, Fig. 4). The Mollusca class, the Sphaeriidae family and the *Sphaerium* sp. genus had significant IndVal values above 80% for stream 3 as well as the *Oulimnius* sp. genus and the corresponding Elmidae family (Tables 6–8). Odonata, in general (Table 6, Fig. 4), and the damselfly family Coenagrionidae in particular (Table 7), were characteristic of stream 3, although no specific odonatan genus contributed significantly (Table 8). On the other hand, the coleopteran genus *Stictonectes* sp. and the dipteran genera *Polypedilum* sp. and *Cricotopus* spp. were significantly characteristic of stream 3, while higher corresponding taxonomic levels such as the orders Coleoptera and Diptera or the families Dytiscidae and Chironomidae were not characteristic due to high dispersion across all five headstreams.

Less than 10% of total taxa were exclusive to streams 1, 2, 4 and 5, corresponding to less than 1% of total captures (Table 5). Very few taxa were significantly characteristic of these streams, namely the family Limnephilidae, the genera *Stenophylax* sp. and *Brillia* sp. in stream 5 and the genera *Hydraena* sp. and *Diamasa* sp. in stream 4 (Tables 5 & 6). Taxa common to all five streams accounted only for 12.4% (16 taxa) of total taxon richness, but represented 57.7% of total abundance (Table 5).

Macroinvertebrate temporal patterns

Average macroinvertebrate genus richness per sampling campaign was relatively constant through time (Fig. 7), with the highest values registered during the driest months (July and September), although a single stream was sampled in July. When considering the cumulative taxa for each sampling campaign, June, September and November presented the highest genus richness (Fig. 7, bottom scheme) and a high turnover rate of invertebrate species was observed between sampling occasions. Streams 2, 4 and 5 showed the smallest variations through time. The difference between the richest and the poorest sampling occasion for these streams never exceeded seven taxa. Streams 3 and 1 showed higher differences on

Table 6 Indicator value scores (IndVal) and associated significance (*P*) obtained by Monte Carlo permutations for major groups of macroinvertebrates (mostly orders) in the five studied streams. Taxa are listed in descending order of IndVal, in each group. Significant characteristic taxa of each stream have an IndVal >25 and *P* < 0.05

Taxon	IndVal	<i>P</i> -value	Stream	Abbr.
Coleoptera	61.85	NS	1	Col
Diptera	27.91	NS	1	Dip
Oligochaeta	43.93	NS	2	Oli
Lepidoptera	20.00	NS	2	Lep
Hydracarina	10.91	NS	2	Hyd
Hirudinea	100.00	<0.05	3	Hir
Heteroptera	96.71	<0.05	3	Het
Ephemeroptera	87.93	<0.05	3	Eph
Mollusca	83.33	<0.05	3	Mol
Odonata	66.67	<0.05	3	Odo
Tricladida	33.33	NS	4	Tric
Trichoptera	43.15	NS	5	Tri
Plecoptera	34.75	NS	5	Ple

NS, non-significant Monte Carlo test.

Table 5 Number of taxa (mostly genera) exclusive to each stream and common to all streams. The relative abundance of exclusive taxa in each stream is indicated in brackets in the last column. Others include Lepidoptera, Hydracarina, Tricladida, Hirudinea and Oligochaeta

Taxa	Dip	Ple	Col	Eph	Tri	Het	Odo	Mol	Others	Total (%)
Exclusive to										
Stream 1	8		2	1						11 (0.17)
Stream 2	5		1		1				1	8 (0.13)
Stream 3	7	1	11	3	2	4	4	2	2	36 (3.70)
Stream 4	5		2	1	2	1			1	12 (0.35)
Stream 5	4	1	1		1					7 (0.83)
Common to all streams	8	1	3	2	1				1	16 (57.7)

Taxon abbreviations are defined in Table 6.

Table 7 Indicator value scores (IndVal) and associated significance (*P*) obtained by Monte Carlo permutations for macroinvertebrate families in the five studied streams (See caption of Table 6 for further information)

Taxon	IndVal	<i>P</i> -value	Stream	Abbr.	Taxon	IndVal	<i>P</i> -value	Stream	Abbr.
Culicidae (Dip.)	38.61	NS	1	Cul	Baetidae (Eph.)	21.88	NS	3	Bae
Empididae (Dip.)	33.56	NS	1	Emp	Lestidae (Odo.)	16.67	NS	3	Les
Helophoridae (Col.)	33.28	NS	1	Hel	Aeshnidae (Odo.)	16.67	NS	3	Aes
Anthomyiidae (Dip.)	30.91	NS	1	Ant	Gerridae (Het.)	16.67	NS	3	Ger
Chironomidae (Dip.)	27.81	NS	1	Chi	Hydrochidae (Col.)	16.67	NS	3	Hydc
Hydrophilidae (Col.)	17.75	NS	1	Hyd	Dryopidae (Col.)	16.67	NS	3	Dry
Rhagionidae (Dip.)	10.00	NS	1	Rha	Scirtidae (Col.)	16.67	NS	3	Sci
Oligochaeta	43.93	NS	2	Oli	Psychodidae (Dip.)	16.67	NS	3	Psy
Tipulidae (Dip.)	20.00	NS	2	Tip	Ferissidae (Mol.)	16.67	NS	3	Fer
Pyrallidae (Lep.)	20.00	NS	2	Pyr	Hydraenidae (Col.)	56.14	IS	4	Hydd
Hydracarina	10.91	NS	2	Hid	Chloroperlidae (Ple.)	52.78	IS	4	Chl
Leptophlebiidae (Eph.)	100.00	<0.05	3	Lep	Rhyacophilidae (Tri.)	44.44	IS	4	Rhy
Erpobdellidae (Hir.)	100.00	<0.05	3	Erp	Mesoveliidae (Het.)	33.33	NS	4	Mes
Elmidae (Col.)	88.90	<0.05	3	Elm	Lepidostomatidae (Tri.)	33.33	IS	4	Lem
Sphaeriidae (Mol.)	83.33	<0.05	3	Sph	Ephydriidae (Dip.)	33.33	NS	4	Eph
Coenagrionidae (Odo.)	66.67	<0.05	3	Coe	Tricladida	33.33	IS	4	Tric
Corixidae (Het.)	66.67	<0.05	3	Cor	Leuctridae (Ple.)	31.14	NS	4	Leu
Polycentropodidae (Tri.)	62.07	<0.05	3	Pol	Simuliidae (Dip.)	22.00	NS	4	Sim
Notonectidae (Het.)	44.12	NS	3	Not	Philopotomidae (Tri.)	18.21	NS	4	Phi
Siphonuridae (Eph.)	43.37	NS	3	Sip	Limnephilidae (Tri.)	78.01	<0.05	5	Lim
Dytiscidae (Col.)	42.89	NS	3	Dyt	Nemouridae (Ple.)	34.19	NS	5	Nem
Perlodidae (Ple.)	33.33	NS	3	Per	Curculionidae (Col.)	33.33	IS	5	Cur
Leptoceridae (Tri.)	33.33	NS	3	Lec	Stratiomyidae (Dip.)	33.33	IS	5	Str
Ceratopogonidae (Dip.)	24.56	NS	3	Cer	Limoniidae (Dip.)	22.65	NS	5	Lio

NS, non-significant Monte Carlo test; IS, insufficient data to test for significance.

Taxon abbreviations in brackets are defined in Table 6.

genus richness between sampling occasions, varying in 20 and 12 taxa, respectively. Average macroinvertebrate abundance showed some variations, with highest numbers recorded in June and November and lowest during August (Fig. 7).

No significant differences were found when comparing diversity metrics [abundance (N) and taxon richness (S)] among flow conditions, connected pools and isolated pools (GLS-based ANOVA: $F_S = 0.15$, d.f. = 2, 15, $P = 0.86$; $F_N = 0.07$, d.f. = 2, 15, $P = 0.93$), independently of the stream, nor when all pools were considered as a single category (GLS-based ANOVA: $F_S = 0.34$, d.f. = 1, 16, $P = 0.57$; $F_N = 0.00$, d.f. = 1, 16, $P = 0.99$).

Among the most frequent and abundant genera (Fig. 4), several distinct temporal patterns were observed such as: (i) genera that were captured only during the driest months, like Odonata, (ii) genera that were captured only before the driest months but never later, like the mayfly *Siphonurus* sp., (iii) genera mainly observed during first stages of colonization,

like the chironomid *Krenosmittia* sp. and the caddisfly *Stenophylax* sp., (iv) genera with high frequency of occurrence and abundance such as the chironomids *Corynoneura* sp. and *Eukiefferiella* sp. and the stonefly *Nemoura* sp.

Streams showed different temporal patterns concerning the presence of invertebrates characteristic of lotic (EPT) and lentic habitats (OCH) and related to extreme conditions (Diptera and Oligochaeta) (Fig. 8). Stream 1 was highly consistent through time, with high abundance of tolerant taxa, mostly Diptera, and few EPT genera (Fig. 8). High abundance of tolerant taxa (oligochaetes and dipterans) was also very constant among sampling occasions in stream 2, but with several EPT genera, which slightly decreased during summer months (August and September). Streams 4 and 5, with a longer dry period, presented a recolonization phase (November) somewhat different from the remaining sampling occasions, mainly due to a decrease in the abundance of tolerant species (Diptera), the dominance of *Nemoura* sp. and the

Table 8 Indicator value scores (IndVal) and associated significance (*P*) obtained by Monte Carlo permutations for macroinvertebrate taxa (mostly genera) in the five studied streams (see caption of Table 6 for further information). Only taxa with IndVal >25 are shown

Taxon	IndVal	<i>P</i> -value	Stream	Abbr.	Taxon	IndVal	<i>P</i> -value	Stream	Abbr.
<i>Tanytarsus</i> spp. (Dip.)	55.68	NS	1	Tan	<i>Isoperla</i> sp. (Plec.)	33.33	NS	3	
<i>Macropelopia</i> sp. (Dip.)	49.33	NS	1	Mac	<i>Coenagrion</i> sp. (Odo.)	33.33	NS	3	
<i>Agabus</i> sp. (Col.)	48.95	NS	1	Aga	<i>Athripsodes</i> sp. (Tri.)	33.33	NS	3	
<i>Hydroporus</i> sp. (Col.)	39.74	NS	1	Hydp	<i>Larsia</i> sp. (Dip.)	33.33	NS	3	
Culicidae (Dip.)	38.61	NS	1	Cul	<i>Paratendipes</i> sp. (Dip.)	33.33	NS	3	
<i>Hydrobaenus</i> sp. (Dip.)	36.44	NS	1	Hyd	<i>Microtendipes</i> sp. (Dip.)	33.33	NS	3	
<i>Helophorus</i> sp. (Col.)	33.28	NS	1	Hel	<i>Eukiefferiella</i> spp. (Dip.)	28.67	NS	3	Euk
<i>Enochrus</i> sp. (Col.)	32.73	NS	1		<i>Hydraena</i> sp. (Col.)	66.67	<0.05	4	
<i>Helochaers</i> sp. (Col.)	32.73	NS	1		<i>Diamesa</i> sp. (Dip.)	66.67	<0.05	4	
<i>Bidessus</i> sp. (Col.)	32.41	NS	1		<i>Chloroperla</i> sp. (Plec.)	52.78	IS	4	
<i>Psectrocladius</i> sp. (Dip.)	26.31	NS	1	Pse	<i>Baetis</i> sp. (Eph.)	33.33	NS	4	
<i>Zavrelimyia</i> sp. (Dip.)	62.52	NS	2	Zav	<i>Mesovelgia</i> sp. (Het.)	33.33	IS	4	
Oligochaeta	43.93	NS	2	Oli	<i>Coelostoma</i> sp. (Col.)	33.33	IS	4	
<i>Chaetocladius</i> spp. (Dip.)	36.22	NS	2	Cha	<i>Esolus</i> sp. (Col.)	33.33	NS	4	
<i>Procladius</i> sp. (Dip.)	27.16	NS	2	Pro	<i>Prosrhyacophila</i> sp. (Tri.)	33.33	NS	4	
<i>Dina</i> sp. (Hir.)	100.00	<0.05	3		<i>Lepidostoma hirtum</i> . (Tri.)	33.33	NS	4	
<i>Sphaerium</i> sp. (Mol.)	83.33	<0.05	3		<i>Heleniella</i> spp. (Dip.)	33.33	IS	4	
<i>Oulimnius</i> sp. (Col.)	82.06	<0.05	3	Oul	Ephydriidae (Dip.)	33.33	NS	4	
<i>Polypedilum</i> spp. (Dip.)	81.73	<0.05	3	Pol	<i>Polycelis</i> sp. (Tricladida)	33.33	NS	4	
<i>Habrophlebia</i> sp. (Eph.)	66.67	<0.05	3	Hap	<i>Leuctra</i> sp. (Plec.)	32.79	NS	4	Leu
<i>Hesperocorixa</i> sp. (Het.)	66.67	<0.05	3		<i>Stenophylax</i> sp. (Tri.)	76.49	<0.05	5	Ste
<i>Stictonectes</i> sp. (Col.)	66.67	<0.05	3		<i>Brillia</i> sp. (Dip.)	66.67	<0.05	5	Bri
<i>Cricotopus</i> spp. (Dip.)	59.58	<0.05	3	Cri	<i>Heterotrissicladus</i> sp. (Dip.)	56.89	NS	5	Het
<i>Plectrocnemia</i> sp. (Tri.)	59.18	<0.05	3	Ple	<i>Parametriocnemus</i> sp. (Dip.)	34.60	NS	5	Parc
<i>Toetenia</i> spp. (Dip.)	52.28	NS	3	Tve	<i>Nemoura</i> sp. (Plec.)	34.36	NS	5	Nem
<i>Ischnura</i> sp. (Odo.)	50.00	NS	3		<i>Chaetopteryx</i> sp. (Tri.)	33.33	NS	5	
<i>Ablabesmyia</i> sp. (Dip.)	50.00	IS	3		<i>Pseudorthocladus</i> sp. (Dip.)	33.33	NS	5	
<i>Notonecta</i> sp. (Het.)	44.12	NS	3		<i>Prodiamesa</i> sp. (Dip.)	33.33	NS	5	
<i>Siphonurus</i> sp. (Eph.)	43.37	NS	3	Sip	Stratiomyidae (Dip.)	33.33	NS	5	
<i>Corynoneura</i> spp. (Dip.)	42.51	NS	3	Cor	<i>Micropsectra</i> spp. (Dip.)	31.81	NS	5	Mic
<i>Thienemanniella</i> spp. (Dip.)	40.82	NS	3	Thi	<i>Krenosmittia</i> sp. (Dip.)	31.17	NS	5	Kre
<i>Paratrachocladus</i> sp. (Dip.)	34.45	NS	3	Part	<i>Allogamus</i> sp. (Tri.)	29.17	NS	5	

NS, non-significant Monte Carlo test; IS, insufficient data to test for significance.

Taxon abbreviations in brackets are defined in Table 6.

presence of a chironomid species, *Krenosmittia* sp. (Fig. 4). Stream 3 had fewer EPT genera and more OCH genera during summer months (July, August and September) and tolerant taxa were mainly present during spring months.

Discussion

Spatial and temporal macroinvertebrate abundance

Although the studied high altitude streams were anthropogenically undisturbed systems (no nutrient enrichment or obvious alterations of hydromorphological parameters), mountain streams are strongly influenced by natural disturbances related to wind,

temperature and hydrological conditions. Hydrological changes were the most obvious natural disturbances identified in the present study, mainly related to wet, cold winters and early springs and dry, warm summers. Drought, as an unpredictable (or predictable; see Lake, 2003) low flow period with unusual duration, extent, severity or intensity (Humphries & Baldwin, 2003), leads to a loss of longitudinal and lateral hydrological connectivity, which causes direct or indirect changes in stream ecosystems properties (Bonada *et al.*, 2006). On the other hand, high flows, although restoring the connectivity, are more sudden events leading to the displacement of benthic communities (e.g. Quinn & Hickey, 1990; Chaves *et al.*, 2005). These frequent disturbances in the studied

streams induced the dominance of highly mobile, pioneer specimens such as the family Chironomidae, as also verified in other temporary (Acuña *et al.*, 2005) and high altitude streams (Füreder *et al.*, 2001; Mai-

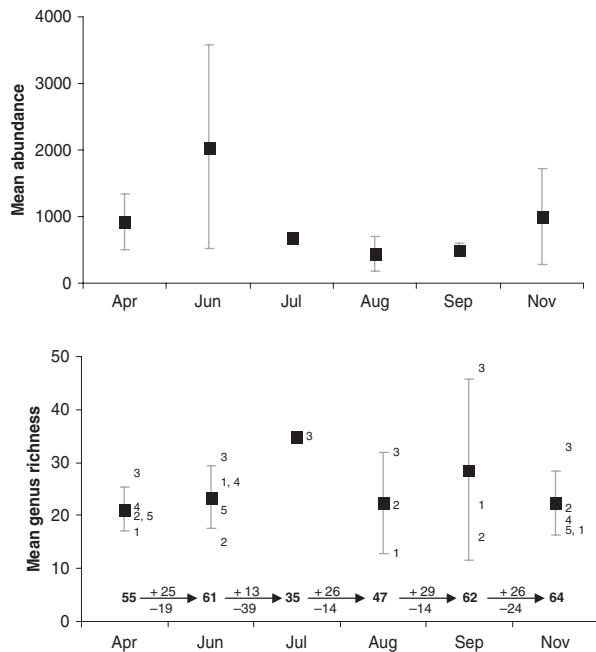


Fig. 7 Mean macroinvertebrate abundance and genus richness on each sampling occasion with respective standard deviations. Genus richness for each stream is also indicated. Total genus richness on each sampling occasion and number of arriving and departing genera between sampling occasions is given on the bottom diagram.

olini & Lencioni, 2001; Maiolini *et al.*, 2006). Chironomidae displayed a relative abundance above 70%, very similar to what was found in the Alpine Italian inlets (Maiolini *et al.*, 2006) and glacial tributaries at upper altitudes (Maiolini & Lencioni, 2001) but considerably higher than in most undisturbed streams. Although chironomids are commonly the most ubiquitous and dominant insect family in most freshwater systems (Williams & Feltmate, 1992; Cranston, 1995), in perennial undisturbed stream sites located in the Mondego River basin at slightly lower altitudes (950–1060 m) chironomid abundance never exceeded 50% of total abundance (e.g. sites 1 and 2 in Chaves *et al.*, 2006).

Similar to previous studies, Plecoptera was the subdominant group in the present study (Maiolini & Lencioni, 2001). Nemouridae and Limnephilidae were the best represented stonefly and caddisfly families, respectively, as also described by Maiolini *et al.* (2006) but, while these authors referred to the sporadic presence of Odonata, Heteroptera and Coleoptera, here coleopterans were found in every stream and on almost every sampling occasion, and sometimes exceeded the abundance of Ephemeroptera. This last group is also known to be poorly represented on isolated mountaintops, particularly above the tree line (Williams & Feltmate, 1992) and, as expected, the non-insects accounted for a small percentage of total abundance there (Williams, 1987; Maiolini *et al.*, 2006). Moreover, the abundance of major invertebrate

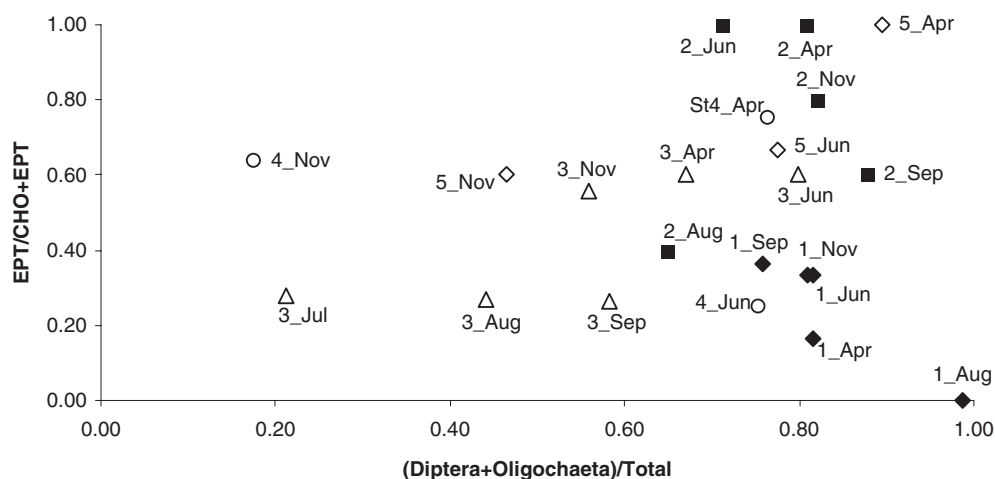


Fig. 8 Relationships between Ephemeroptera, Plecoptera, and Trichoptera (EPT) richness relative to EPT + Coleoptera, Heteroptera and Odonata (OCH) richness [EPT/(EPT + OCH)] and Diptera + Oligochaeta abundances relative to total captures [Diptera + Oligochaeta/Total] for each stream and sampling occasion.

groups in these streams at altitudes between 1400 m a.s.l. and 1700 m a.s.l. seems to be similar to that found in the Italian Alps above 2000 m a.s.l. (Maiolini *et al.*, 2006) and glacial streams 1 km away from the snout of the glacier and already with some krenal influence (Maiolini & Lencioni, 2001).

Over time, the greatest abundances found in these high altitude intermittent systems were also related to flow variations, as previously shown by Boulton & Lake (1992). The first and larger peak occurred in late spring as flow diminished and the second in autumn when flow stabilized after the first heavy rains. These were expected results since increasing food availability (detritus) during the flow reduction (or stabilization) can lead to a considerable increase in invertebrate abundance (Boulton & Lake, 1992; Acuña *et al.*, 2005). These peaks were especially related to the appearance or increasing numbers of Chironomidae genera as described by Acuña *et al.* (2005) and Boulton & Lake (1992), but also of *Habrophlebia* sp., *Siphonurus* sp. and *Helophorus* sp. The abundance peak of late spring dropped rapidly in response to flow cessation (habitat contraction) and probable consequent predation pressure (Acuña *et al.*, 2005) during summer months, when Tanyptodinae and some Odonata and Heteroptera species started to arrive. High abundance variation within streams was probably due to alterations in flow conditions. The smaller number of samples collected in streams 4 and 5 and the similar abiotic conditions between sampling occasions in these streams (dry in summer) may explain the highest homogeneity in their abundance values. On the other hand, the bedrock substrata of streams 1, 2 and 3 allowed the formation of long lasting pools with consequent high macroinvertebrate abundance variations.

Taxon richness of pools and flow conditions

Ward, Tockner & Schiemer (1999) suggested that low genus richness should be associated with the loss of connectivity during the warmest months since exchanges of matter, energy and organisms are confined to isolated pools. However, maximum genus richness for streams that showed a marked variation through time (streams 1 and 3) occurred in the connected pools or in isolated pools shortly after flow stopped. This indicates the occurrence of invertebrate migration from the drying riffles to the pools, simi-

larly to what Williams & Hynes (1977) and Boulton & Lake (1992) have found. To further develop this hypothesis, Acuña *et al.* (2005) suggested that flow reduction increased the detritus coverage, thereby enhancing spatial habitat heterogeneity, which may increase taxon richness.

Typical lentic habitat taxa, such as OCH, were detected mainly during summer months, particularly Odonata species, but also some Heteroptera (e.g. *Hesperocorixa* sp., *Sigara* sp.) and Coleoptera genera (*Hydrochus* sp., *Stictometes* sp.) that arrived between the June and July sampling occasions. This is typically found in Mediterranean streams (Rieradevall *et al.*, 1999). All Odonata and most Heteroptera species were detected only in stream 3. The overlap between benthic composition of pools and flow conditions seems to be greater in more temporary streams, whereas habitat specific assemblages occurred in the most permanent stream (i.e. stream 3), similar to what Boulton & Lake (1992) have described. Even so, the macroinvertebrate taxonomic composition of isolated pools, connected pools or flowing conditions was not significantly distinct, similar to Rabeni *et al.* (2002) but in contrast to most macrohabitat connectivity studies performed in Mediterranean-climate regions (e.g. Rieradevall *et al.*, 1999; Bonada *et al.*, 2006). This difference can be related to the influence of the mountain environment, which might complicate the dispersion ability of OCH taxa, facilitating the dominance of chironomids, and simultaneously ameliorating physicochemical conditions in pools. Pools were very shallow systems influenced by windy conditions, where dissolved oxygen never dropped below 6 mg L⁻¹, enabling the presence of oxygen sensitive macroinvertebrate taxa (*Habrophlebia* sp.) even during the driest month.

The cumulative family richness found in the earlier mentioned perennial streams (sites 1 and 2 in Chaves *et al.*, 2006) was higher than the present study, with a disparity of nine families in relation to stream 3 (the most permanent one, although forming pools) and 19–25 families when compared to the other studied streams. The loss of taxon richness with increasing altitude (e.g. Maiolini & Lencioni, 2001) and with the incidence of dry periods (e.g. Williams, 1987) was as expected. Genus richness in the present study was slightly higher than that found in Alpine streams, although genus composition was somewhat different (Maiolini & Lencioni, 2001).

We found considerable overlap between macroinvertebrate faunas of the studied streams, probably due to spatial proximity and similitude between flow regimes (all streams dry upto form pools), although only 16 taxa (57.7% abundance) were common to all five streams, half of which belonging to the order Diptera (36.1% abundance). These common taxa included different taxonomic levels (class, family and genus), so it is possible that if all specimens had been identified to species or genus the similarity between streams would decrease even more.

Length of the dry period and environmental gradients

Temporary streams are essentially natural waterbodies that experience a recurrent dry period of varying duration (e.g. Williams, 1987, 2006). The three temporary flow regimes studied included streams that (i) never dried but formed pools, (ii) dried for less than 3 months and (iii) dried for at least 3 months. The most permanent stream (stream 3), showed the most dissimilar macroinvertebrate community with the highest richness, diversity, EPT and OCH genus richness, more characteristic taxa identified by the IndVal procedure and distinct community structure as showed by CCA ordinations. Unsurprisingly, the occurrence of a dry period clearly disturbs the aquatic organisms living there (Williams, 1987; Delucchi, 1988; Rieradevall *et al.*, 1999; Boulton, 2003) and, further, species richness apparently increases with increasing water permanence (Williams, 1987, 2006). In accordance with this, abiotic gradients established along first ordination canonical axes always related to flow variations, which are among factors best related to benthic community composition in running waters (e.g. Statzner & Higler, 1986; Sandin & Johnson, 2004). Streams with lower discharge conditions and therefore a higher probability of drying were clearly separated from the most permanent stream along the first ordination axes. No major differences were found for the flow regimes with different length of the dry period, with the exception of the recolonization period (November). Streams that dried for a longer period of time (4 and 5) presented a distinct recolonization phase, while streams 1 and 2 had similar benthic community structures even after the dry period. The absence of solid rock substrata, in contrast to streams 1, 2 and 3, and the dominant gravel sediment in streams 4 and 5 probably enabled the colonization by

some stonefly species such as *Nemoura* sp. (Williams, 1987) and chironomid species such as *Krenosmittia* sp. (Ferrington, 1984) from the hyporheos.

Taxonomic identification levels

The choice of taxonomic level (genus, family and order) did not seem critical in identifying environmental gradients and streams with the highest macroinvertebrate richness and diversity. However, the characteristic taxa of streams (with specimens present in more than 50% of the samples from one group and with a relative abundance in that same group of more than 50%), established by the IndVal procedure, varied considerably with the taxonomic level. A macroinvertebrate order could be characteristic of a site but by enclosing so many different families and genera that occurred on different sampling occasions none of these lower taxa were characteristic of that site. On the other hand, some particular families or genera were characteristic of a specific stream, although higher taxonomic levels were not characteristic of a site due to high dispersion among streams.

The need for pre-impact long term data sets

In the future, global climate changes may transform perfectly predictable seasonal droughts, typical from Mediterranean climate, into supra-seasonal droughts leading to disturbances that increase in strength with time (see Lake, 2003). Subsequently, changes in water quantity and quality may also be expected (Allan & Flecker, 1993) and the headwater of mid-latitude streams will, at least, be reduced in extent if warmer climates prevail (Regier & Meisner, 1990). If predictions are true, warmer winters, with increasing rain but decreasing snow deposition, and dryer summers are expected for high mountain scenarios. These changes may induce a reduction in streams spring and summer flows and an increase in the duration of the dry period, leading to a decline in the riverine fauna biodiversity, which cannot be easily counteracted by conservation measures. It is important to obtain consistent pre-impact long-term data sets in order to incorporate the natural variability of these systems and accurately predict the ecological consequences of future climate changes in high altitude scenarios.

Although data analyses helped summarize important information, complex ecological patterns were found, probably deriving from multiple and simultaneous interactions of variables related to flow intermittency and hydrologic permanence in headwaters, high altitude, different vegetation covers and river bed substrata. All these factors should be carefully considered to avoid misinterpretation when evaluating the stream condition. It was difficult to find close faunal similarities among the studied streams, despite their spatial proximity, so reference conditions establishment for these high altitude streams with siliceous geology and very small catchment areas (<10 km²) should also imply long-term data collection and more detailed physical characterization.

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Appendix Distribution of macroinvertebrate taxa in five high altitude streams of Serra da Estrela (Portugal)

Taxa	Author	Stream				
		1	2	3	4	5
Tricladida	<i>Polycelis</i> sp.	Ehrenberg, 1831			*	
Oligochaeta		***	***	***	*	**
Hirudinea						
Erpobdellidae	<i>Erpobdella</i> sp.	de Blainville, 1818		*		
	<i>Dina</i> sp.	Blanchard, 1892		**		
Mollusca						
Ferrissidae	<i>Ferrissia</i> sp.	Fullaway, 1923		*		
Sphaeriidae	<i>Sphaerium</i> sp.	Scopoli, 1777		**		
Hydracarina			*	*		
Ephemeroptera						
Leptophlebiidae				**		
	<i>Habrophlebia</i> sp.	Eaton, 1881		***		
Baetidae		**	*	**	*	*
	<i>Proclon bifidu</i>	(Bengtsson, 1912)		**		
	<i>Cloeon</i> sp.	Leach, 1815	*			
	<i>Baetis</i> sp.	Leach, 1815			*	
Siphonuridae	<i>Siphonurus</i> sp.	Eaton, 1868	*	**	***	*
Plecoptera						
Nemouridae	<i>Nemoura</i> sp.	Latreille, 1796	***	***	***	***
	<i>Protonemura</i> sp.	Kempny, 1898	*	*		
Leuctridae						*
	<i>Leuctra</i> sp.	Stephens, 1836	**	*	**	**
Perlodidae	<i>Isoperla</i> sp.	Banks, 1906		*		
Chloroperlidae	<i>Chloroperla</i> sp.	Newman, 1836			**	*

Appendix (Continued)

			Stream				
Taxa		Author	1	2	3	4	5
Trichoptera							
Philopotamidae	<i>Womaldia</i> sp.	McLachlan, 1865		**		**	
Polycentropodidae	<i>Plectrocnemia</i> sp.	Stephens, 1836		**	**	*	
	<i>Cyrnus</i> sp.	Stephens, 1836			*		
Rhyacophilidae	<i>Pararhyacophila</i> sp.	†			*	*	*
	<i>Prosrhyacophila</i> sp.	†				*	
Limnephilidae	<i>Stenophylax</i> sp.	Kolenati, 1848	*	*	**	*	**
	<i>Grammotaulius</i> sp.	Kolenati, 1848		*			
	<i>Allogamus</i> sp.	Schmid, 1955			*		*
	<i>Chaetopteryx</i> sp.	Stephens, 1829					*
Lepidostomatidae	<i>Lepidostoma hirtum</i>	(Fabricius, 1775)				*	
Leptoceridae	<i>Athripsodes</i> sp.	Billberg, 1820			*		
Diptera							
Psychodidae					*		
Culicidae			***	*	*		
Simuliidae	<i>Simulium</i> sp.	Latreille, 1802	**	**	**	**	**
	<i>Prosimulium</i> sp.	Roubaud, 1906	*				
Ceratopogonidae			**	*	**		
Chironomidae							
Tanypodinae	<i>Ablabesmyia</i> sp.	Johannsen, 1905			**		
	<i>Krenopelopia</i>	Fittkau, 1962		*			
	<i>Larsia</i> sp.	Fittkau, 1962			*		
	<i>Macropelopia</i> sp.	Thienemann, 1916	***	***	**	**	**
	<i>Natarsia</i> sp.	Fittkau, 1962			**		
	<i>Paramerina</i> sp.	Fittkau, 1962	***		**		
	<i>Procladius</i> sp.	Skuse, 1889	***	***	*		
	<i>Trissopelopia</i> sp.	Kieffer, 1923			*		
	<i>Zavrelimyia</i> sp.	Fittkau, 1962	***	***	**		**
Diamesinae	<i>Diamesa</i> sp.	Meigen, 1835				**	
	<i>Pseudokiefferiella</i> sp.	Zavrel, 1941	*				
Prodiamesinae	<i>Prodiamesa</i> sp.	Kieffer, 1906					*
Orthocladiinae	<i>Brillia</i> sp.	Kieffer, 1913					***
	<i>Bryophaenocladus</i> sp.	Thienemann, 1934	*				
	<i>Chaetocladius</i> spp.	Kieffer, 1911	***	***	**	**	
	<i>Corynoneura</i> spp.	Winnertz, 1846	***	***	***	**	***
	<i>Cricotopus</i> spp.	van der Wulp, 1874			***	**	
	<i>Eukiefferiella</i> spp.	Thienemann, 1926	***	***	***	***	***
	<i>Heleniella</i> spp.	Gouin, 1943				*	
	<i>Heterotrissocladius</i> sp.	Spärk, 1923	***	***	***	**	****
	<i>Hydrobaenus</i> sp.	Fries, 1830	**		***		
	<i>Krenosmittia</i> sp.	Thienemann & Krüger, 1939		*		**	***
	<i>Lymnophyes</i> sp.	Eaton, 1875		**	**		**
	<i>Orthocladius</i> sp.	van der Wulp, 1874	**	*	**	*	
	<i>Parametriocnemus</i> sp.	Goetghebuer, 1932	**	***	**	*	***
	<i>Paraphaenocladus</i> sp.	Thienemann, 1924		**	**		
	<i>Paratrachocladius</i> sp.	Santos-Abreu, 1918	**		***		
	<i>Psectrocladius</i> spp.	Wülker, 1956/Kieffer, 1906	**	*	**		
	<i>Pseudorthocladius</i> sp.	Goetghebuer, 1932					**
	<i>Rheocricotopus</i> spp.	Thienemann & Harnisch, 1932		**	*	**	
	<i>Smittia</i> sp.	Holmgren, 1869		*			
	<i>Stilocladius</i> spp.	Rossaro, 1979		**			
	<i>Thienemanniella</i> spp.	Kieffer, 1911		**	***	**	**
	<i>Toetenia</i> spp.	Kieffer, 1922	***	**	****		**

Appendix (Continued)

Taxa	Author	Stream				
		1	2	3	4	5
Chironomini	<i>Chironomus</i> sp.	Meigen, 1803	*			
	<i>Microtendipes</i> sp.	Kieffer, 1915		*		
	<i>Paratendipes</i> sp.	Kieffer, 1911		**		
	<i>Polypedilum</i> spp.	Kieffer, 1912		***		*
Tanytarsini	<i>Micropsectra</i> spp.	Kieffer, 1909	***	**	**	***
	<i>Paratanytarsus</i> sp.	Thienemann & Bause, 1913	*			
	<i>Stempellinella</i> sp.	Brundin, 1947		**	*	
	<i>Tanytarsus</i> spp.	van der Wulp, 1874	****	****	**	**
Tipulidae			*			
Limoniidae			**	***	*	**
Pediciini					*	
Hexatomini					*	
Stratiomyidae						*
Empididae			*			
Hemerodromiinae			*			
	<i>Rhamphomyia</i> sp.	Meigen, 1822	*			*
Rhagionidae	<i>Chrysopilus</i> sp.	Macquart, 1826	*	*		
Ephydriidae					*	
Anthomyidae			**	*		
	<i>Limnophora</i> sp.	Robineau-Desvoidy, 1830	*			
	<i>Lispe</i> sp.	Latreille, 1796	*			
Lepidoptera						
Pyalidae			*			
Coleoptera						
Dytiscidae			**		*	
	<i>Dytiscus</i> sp.	Linnaeus, 1758		*		
	<i>Meladema</i> sp.	Laporte, 1835		*		
	<i>Agabus</i> sp.	Leach, 1817	**	*	**	*
	<i>Stictonectes</i> sp.	Brinck, 1943			**	
	<i>Deronectes</i> sp.	Sharp, 1882			*	
	<i>Bidessus</i> sp.	Sharp, 1882	**		*	
	<i>Hydroporus</i> sp.	Clairville, 1806	**	*	**	**
	<i>Laccornis</i> sp.	Gozis, 1914		*		
	<i>Scarodytes</i> sp.	Gozis, 1914			*	
	<i>Acilius</i> sp.	Leach, 1817			*	
	<i>Ilybius</i> sp.	Erichson, 1832	*	*		
Hydrophilidae	<i>Coelostoma</i> sp.	Brullé, 1835			*	
	<i>Anacaena</i> sp.	Thomson, 1859	*			
	<i>Enochrus</i> sp.	Thomson, 1859	*			
	<i>Helochaeres</i> sp.	Mulsant, 1844	*		*	
Helophoridae	<i>Helophorus</i> sp.	Fabricius, 1775	***	*	**	**
Hydrochidae	<i>Hydrochus</i> sp.	Fallén, 1823			**	
Hydraenidae	<i>Hydraena</i> sp.	Kugelann 1794			*	
	<i>Limnebius</i> sp.	Leach 1815			*	
Dryopidae	<i>Dryops</i> sp.	Olivier, 1791			*	
Elmidae	<i>Esolus</i> sp.	Mulsant & Rey, 1872			**	*
	<i>Oulimnus</i> sp.	Des Gozis, 1886			***	*
	<i>Limnius</i> sp.	Illiger, 1802			*	
	<i>Stenelmis</i> sp.	Dufour, 1835	*			
Scirtidae	<i>Cyphon</i> sp.	Paykull, 1799			*	
Curculionidae						*
Odonata						
Lestidae	<i>Chalcolestes viridis</i>	(Vander Linden, 1825)			*	
Coenagrionidae	<i>Ischnura</i> sp.	Charpentier, 1840			**	

Appendix (Continued)

Taxa		Author	Stream				
			1	2	3	4	5
	<i>Coenagrion</i> sp.	Kirby, 1890			*		
Aeshnidae	<i>Anax</i> sp.	Leach, 1815			*		
Heteroptera							
Notonectidae	<i>Notonecta</i> sp.	Linnaeus, 1758	*		*		
Corixidae	<i>Hesperocorixa</i> sp.	Kirkaldy, 1908			***		
	<i>Sigara</i> sp.	Fabricius, 1775			**		
	<i>Parasigara</i> sp.	Poisson, 1957			*		
Mesoveliidae	<i>Mesovelia</i> sp.	Mulsant & Rey, 1852				*	
Gerridae	<i>Gerris</i> sp.	Fabricius, 1794			*		

Number of * indicates taxon total abundance, with * < 10; ** 10–100; *** 100–1000 and **** > 1000 specimens in a 10 m stream reach.

[†]This sub-genus belongs to the genus *Rhyacophila* Pictet, 1834. Major groups considered in this study are shown in bold.